



Revision Notes: Combinatorics Basics & Prime Numbers

Overview

This class covers combinatorial mathematics focusing on the basics of permutations, combinations, and an introduction to prime numbers using the Sieve of Eratosthenes. The class also explores the concepts of addition and multiplication principles in combinatorics and their applications in problem-solving.

Combinatorics

Addition and Multiplication Rules

- **Addition Rule (OR):** Used when events cannot happen simultaneously. E.g., choosing between two non-overlapping paths.
- **Multiplication Rule (AND):** Used for composite events. Used to compute the total number of ways to perform a series of tasks where each subsequent task does not alter the previous one.

Example:

1. **Route Selection:** Choose a route via different cities to complete a trip. Way counts are added when taking different paths (OR), multiplied when going stepwise (AND) .

Permutations

- **Definition:** Arrangement of objects where the order matters.
- **nPr Formula:** Used when arranging r items out of n items. The formula is: $nPr = \frac{n!}{(n-r)!}$
- **Examples:** Arranging 3 distinct characters from 4 available (ABCD), resulting in different sequences like ABC, ACB, BAC, etc. .



- **Definition:** Selection of items where the order does not matter.
- **nCr Formula:** $nCr = \frac{n!}{r!(n-r)!}$
- **Properties:**
 - Selecting 0 or n items results in exactly one way.
 - Symmetry in selection: $nCr = nC(n-r)$.
 - Recursive relationship: $nCr = (n-1)C(r-1) + (n-1)Cr$.
- **Example:** Selecting teams or groups from a set of players without regard to the order.

Prime Numbers and the Sieve of Eratosthenes

Introduction to Prime Numbers

- **Definition:** Natural numbers greater than 1, having no divisors other than 1 and themselves.
- **Brute Force Approach:** Checking primality by testing divisibility up to the square root of the number, with time complexity $O(n\sqrt{n})$.

Sieve of Eratosthenes

- **Idea:** An efficient algorithm to compute prime numbers up to n by iteratively marking the multiples of each prime number starting from 2.
- **Algorithm Steps:**
 1. Assume all numbers > 1 are prime.
 2. Begin with the first prime (2), iterate and mark its multiples as non-prime.
 3. Move to the next unmarked number, repeating the marking process till the square root of n.
 4. Numbers which remain unmarked are prime.
- **Complexity:**
 - **Time:** $O(n \log \log n)$ due to the consideration of only multiples of primes and iteration up to the square root of n.
 - **Space:** $O(n)$ for storing boolean values representing prime number status.

These notes provide an overview of the fundamental combinatorics and number theory concepts as discussed in the class, with

